

Objections to Gradient Ascent

We've just performed gradient ascent to maximize the revenues of a hypothetical government. Many people who learn gradient ascent have practical if not moral objections to it. Here are some of the arguments that people raise about gradient ascent:

- It's unnecessary because we can do a visual inspection to find the maximum.
- It's unnecessary because we can do repeated guesses, a guess-and-check strategy, to find the maximum.
- It's unnecessary because we can solve the first-order conditions.

Let's consider each of these objections in turn. We discussed visual inspection previously. For our taxation/revenue curve, it's easy to get an approximate idea of the location of a maximum through visual inspection. But visual inspection of a plot does not enable high precision. More importantly, our curve is extremely simple: it can be plotted in two dimensions and obviously has only one maximum on the range that interests us. If you imagine more complex functions, you can start to see why visual inspection is not a satisfactory way to find the maximum value of a function.

For example, consider a multidimensional case. If our economists had concluded that revenue depended not only on tax rates but also on tariff rates, then our curve would have to be drawn in three dimensions, and if it were a complex function, it could be harder to see where the maximum lies. If our economists had created a function that related 10 or 20 or a 100 predictors to expected revenue, it would not be possible to draw a plot of all of them simultaneously given the limitations of our universe, our eyes, and our brains. If we couldn't even draw the tax/revenue curve, then there's no way visual inspection could enable us to find its maximum. Visual inspection works for simple toy examples like the tax/revenue curve, but not for highly complex multidimensional problems. Besides all of that, plotting a curve itself requires calculating the function's value at every single point of interest, so it always takes longer than a well-written algorithm.